MIL-I-85352
12 May 1980
SUPERSEDING
MIL-G-008348
2 July 1976
MIL-G-8348
Class B & C
30 March 1972

MILITARY SPECIFICATION

INFLATOR ASSEMBLY AND GAGE ELEMENTS, PNEUMATIC PRESSURE, REMOTE CONTROL, DIRECT READING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1 SCOPE

1.1 Scope. This specification covers a direct reading, remote control, pneumatic pressure inflator assembly, for use on aircraft tires and struts having pneumatic pressure requirements up to 600 psi. It includes pressure relief provisions to provide for safe inflation Also included are dual chuck stem gages for measuring tire pressure.

1.2 Classification. Inflator assemblies and gage elements shall be as specified in the following table (see 6.2)

PART NUMBER	NOMENCLATURE	PRESSURE RANGE (PSI)
M85352/1	INFLATOR ASSEMBLY KIT	50-600 & 10-150
M85352/2	REMOTE CONTROLLER ASSEMBLY	---
M85352/3-1	GAGE ELEMENT, SIZE 1	50-600
M85352/3-2	GAGE ELEMENT, SIZE 2	10-150
M85352/4-1	DUAL CHUCK STEM GAGE KIT SIZE 1	50-600
M85352/4-2	DUAL CHUCK STEM GAGE KIT SIZE 2	10-150

2 APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Beneficial comments (recommendation, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to ENGINEERING SPECIFICATIONS AND STANDARDS DEPARTMENT (CODE 93), NAVAL AIR ENGINEERING CENTER, LAKEHURST, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6685

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

Federal

QQ-C-320	Chromium Plating (Electrodeposited)
QQ-N-290	Nickel Plating (Electrodeposited)
QQ-P-416	Plating, Cadmium (Electrodeposited)
QQ-Z-325	Zinc Coating, Electrodeposited, Requirements for
PPP-C-1120	Cushioning Material, Uncompressed Bound Fiber for Packaging

Military

MIL-P-116	Preservation, Packaging, Methods of
MIL-D-1000/1	Drawings, Engineering and Associated Data
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-H-13531	Hose, Rubber and Hose Assembly Rubber (Hydraulic Flexible)
MIL-T-27730	Tape, Antiseize, Tetrafluoroethylene, with Dispenser
MIL-I-85352/1	Inflator Assembly Kit
MIL-I-85352/2	Remote Controller Assembly
MIL-I-85352/3	Gage Elements
MIL-I-85352/4	Dual Chuck Stem Gage Kit

STANDARDS

Federal

FED-STD-595	Color

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:50Z
Check the source to verify that this is the currentversion before use.

STANDARDS (Continued)

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-794	Part and Equipment, Procedures for Packaging and Packing of
MIL-STD-808	Finish, Protective, and Codes for Finishing Schemes for Ground and Ground Support Equipment
MIL-STD-831	Test Reports, Preparation of
MIL-STD-889	Dissimilar Metals
MS28889	Valve, Air, High-Pressure Charging
AN912	Bushing, Pipe

(When requesting applicable documents, refer to both title and number. Copies of unclassified documents may be obtained from the Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120. Requests for copies of classified documents should be addressed to the Naval Publications and Forms Center, via the cognizant Government representative.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American National Standard Institute

ANSI B40.1-1974	Gages, Pressure and Vacuum, Indicating Type, Elastic Element

(Copies of the above publication may be obtained from the American National Standards Institute, 10 East 46th Street, New York, New York 10017.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

National Bureau of Standards

Handbook H28 Screw-Thread Standards for Federal Services

(Application for copies shoule be addressed tc the Superintendent of Documents, Government Printing Office, Washington, D C 20402)

3 REQUIREMENTS

3 1 Specification sneets The individual part requirements shall be as specified herein and in accordance with the applicable specification sheets In the event of conflict between the requirements of this specification and the specification sheet, the requirements of the specification sheet shall govern.

3 2 First article The parts furnished under this specification shall be a product which has been inspected and passed the first article inspection specified herein (see 4.3)

3 3 Selection of specifications and standards Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143, except as provided in 3 2 1

3 3 1 Commercial parts Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts such as screws, bolts, nuts, and cotter pins having suitable properties may be used provided

a. They can be replaced by the MS or AN standard parts without alteration

b The corresponding standard part numbers are referenced in the parts list and, if practical, on the contractor's drawings

3 4 Materials. Materials shall conform to applicable specifications or standards and shall be as specified herein. Materials which are not covered by applicable specifications or standards shall be of the best quality, and entirely suitable for the purpose intended

3 4 1 Metals Metals shall be of the corrosion-resistant type or treated to resist corrosion All metal stock used in the fabrication of inflator assemblies shall have a properly refined structure with uniform elemental distribution, sound and free from nonmetallic inclusions, injurious porosity, excess segregation, soft spots, hard spots, and other injurious defects.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:50Z
Check the source to verify that this is the currentversion before use.

3.4.2 Metal parts. All metal parts shall be of corrosion-resistant material or treated in a manner to render them adequately resistant to corrosion.

3.4.3 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.4.4 Fungus proof materials. Materials that are nutrients of fungi shall not be used.

3.5 Design and construction. The inflator assembly shall be designed to provide a safe, reliable and accurate means for remote controlled inflation, deflation and gaging pneumatic pressure. The dual chuck stem gage shall be designed to provide a safe, reliable and accurate means for gaging pneumatic pressure. The inflator assembly and dual chuck stem gages shall be so constructed that parts will not work loose in service. It shall be built to withstand the strains, shocks, vibrations, and environmental conditions incident to shipping, storage, and service. The configurations shall be as shown in Figures 1 and 2. The gage elements Size 1 and 2, shall be complete self-contained units and interchangeable within the inflator assembly and dual chuck stem gage. The relief valve shall provide pressure relief to prevent overinflation.

3.5.1 Threads. All threaded components shall conform to the applicable requirements of the National Bureau of Standards Handbook H-28. All pipe thread connections shall be assembled with antiseize tape conforming to MIL-T-27730.

3.5.2 Fastening devices. All screws, pins, and similar parts shall be installed with adequate means for preventing loss of proper tightness and adjustment. Parts subject to removal or adjustment shall not be swaged, peened, staked, or otherwise deformed.

3.5.3 Welding, brazing, and soldering. Welding, brazing, and soldering shall be performed in accordance with the best commercial practice and shall be used only when such operations are required for assemblies in the original design. In no event shall such processes be used as a repair medium for faulty parts.

3.6 Maintainability. The inflator assembly and dual chuck stem gages shall have a minimum number of parts consistent with reliability. All parts subject to disassembly for maintenance purposes shall be readily accessible for adjustments, repairs, and cleaning without the use of special tools.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

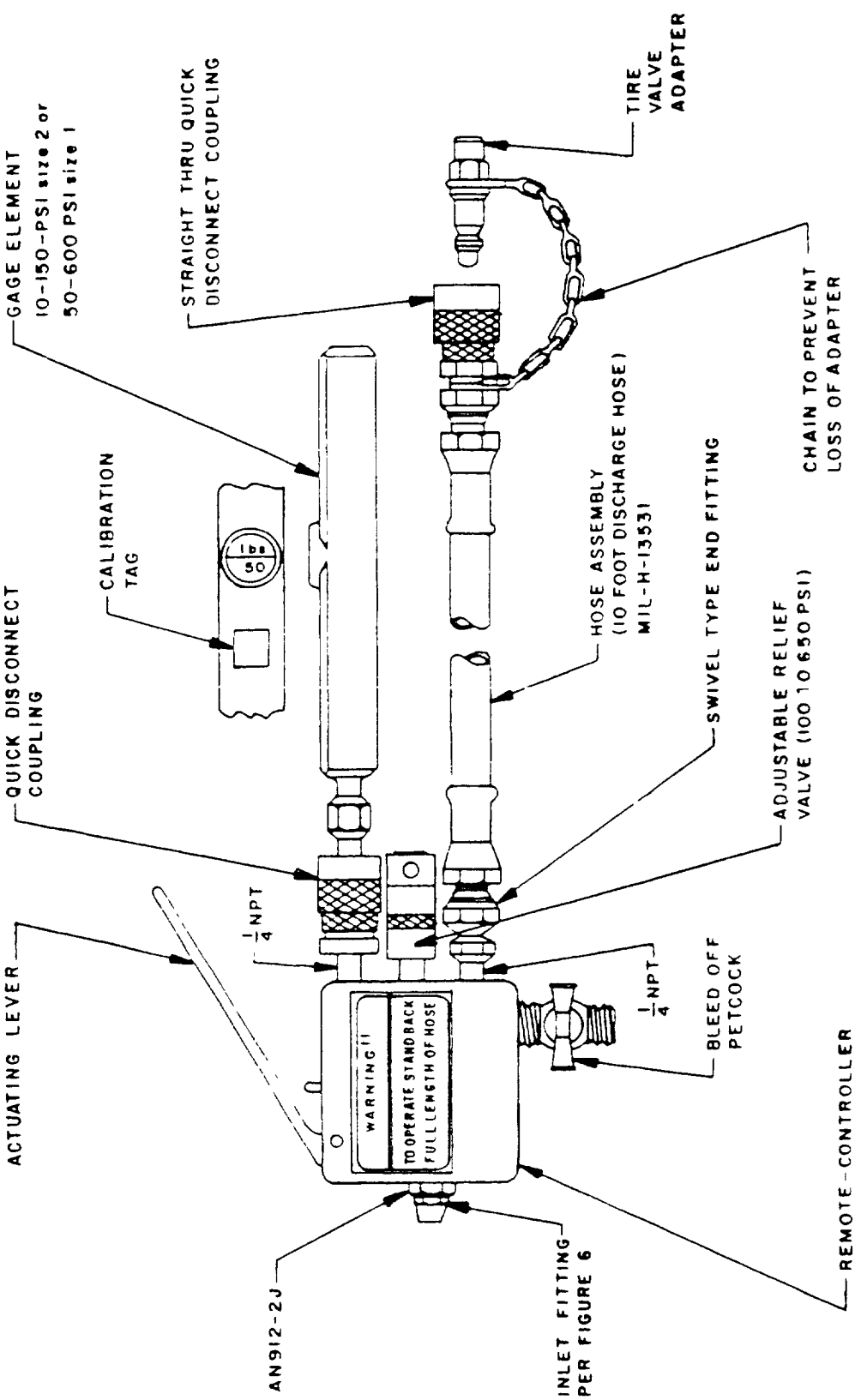


Figure 1. Inflator assembly

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:50Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026/10-26T04:50Z
Check the source to verify that this is the current version before use.

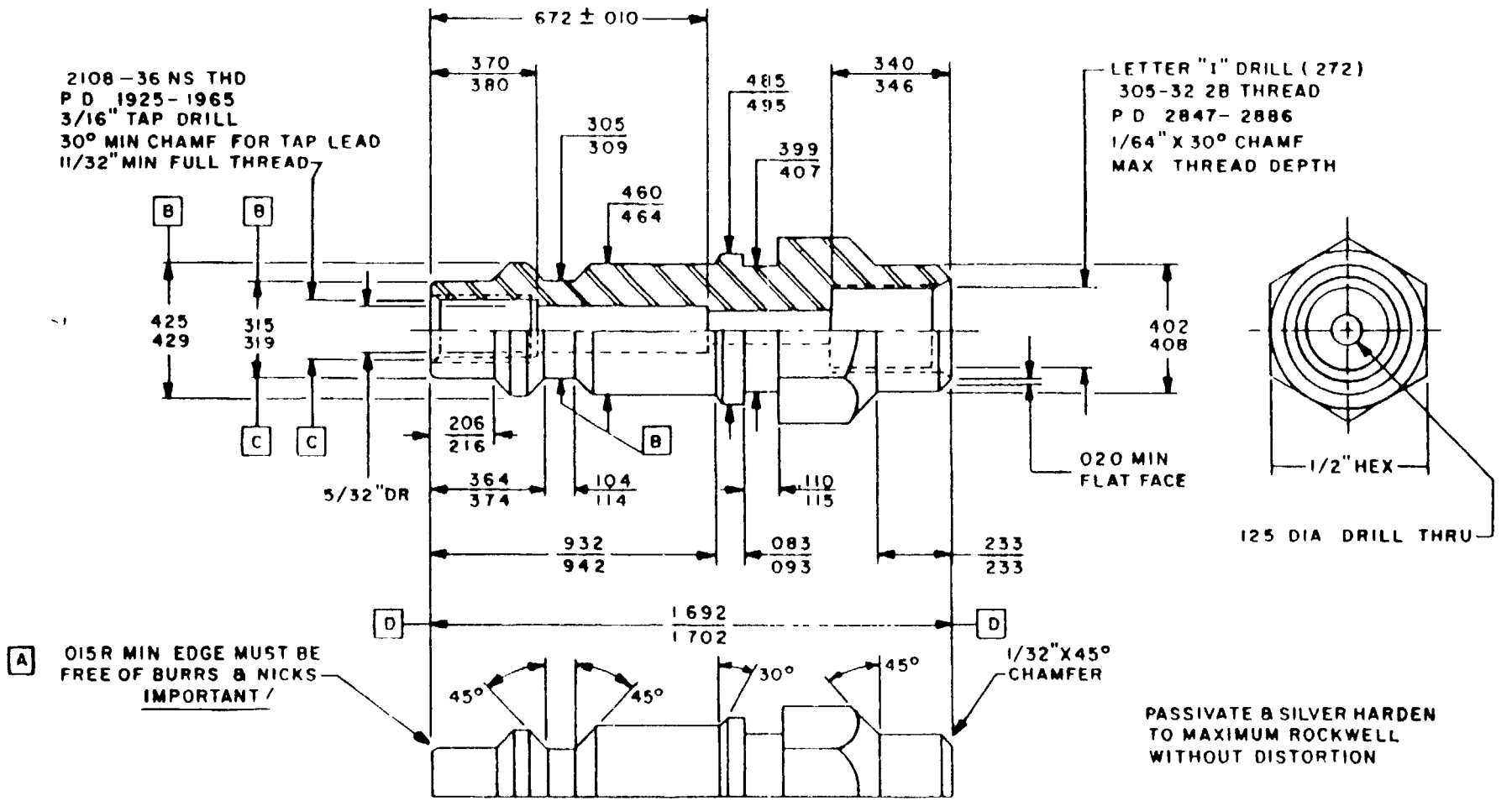


Figure 1 Tire valve adapter

3.7 Performance characteristics. The inflator assembly shall meet the performance tests in Section 4 in the order listed below:

a.	Scale error at room temperature	(4.6.2)
b	Overpressure	(4.6.3)
c	Endurance	(4.6.4)
d.	Drop test	(4.6.5)
e	Cold test	(4.6.6)
f.	Heat test	(4.6.7)
g	Friction test	(4.6.8)
h	Calibration test	(4.6.9)
i	Valve adapter test	(4.6.10)

The dual chuck stem gage shall be subjected to the endurance and drop test only. The gage element used in the dual chuck stem gage shall be tested as part of the inflator assembly for the remaining test listed

3.8 Inflator assembly kit The inflator assembly kit shall conform to MIL-I-85352/1 and consist of a gage element, remote-controller, hose/tire valve adapter assembly, relief valve, carrying case, and inlet fitting.

3.8.1 Gage element The gage element shall conform to MIL-I-85352/3. The gage element shall consist of an indicator (pressure lbs) and actuating mechanism enclosed in a housing fitted with the male portion of a straight thru quick-disconnect coupling and lens The gage element shall be capable of measuring pneumatic pressure from 50 to 600 psi for Size 1, from 10 to 150 psi for Size 2. The accuracy of the gage element shall be ±6 psi for Size 1, and ±2 psi from 10 to 80 psi and ±4 psi from 81 to 150 psi for Size 2, through their entire range at temperatures from -20° to +140°F. The gage element shall be instantly removable and replaceable as a unit to the remote-controller body by use of a straight thru quick-disconnect coupling. Both gage elements, Size 1 and Size 2, are to be interchangeable for use within the inflator assembly.

3.8.1.1 Indicator. The indicator shall be manufactured from nonferrous metal or plastic. The indicator shall be calibrated from 50 to 600 psig in 10-pound increments for Size 1, from 10 to 150 psig in 2-pound increments for Size 2. Adjacent graduations shall be spaced not less than .026 inch apart. Every fifth graduation shall be indicated by its value with numerals not less than .063 inch in height. The graduations and numerals shall be clearly and legibly marked in black on a white background. The graduations and markings are to be arranged on the indicator in such a manner to permit reading through the lens regardless of position of the indicator.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

3.8.1.2 Actuating mechanism. The indicator piston shall be pressure actuated, and balanced by the resistance of a constant rate (compression) spring. Internal stops shall be provided to control extreme positions of the piston to prevent overloading the spring An external means of adjusting the mechanism shall be provided for correcting the indicator reading. The adjustment shall be capable of correcting an indicator error of from -20 psi to -50 psi at any scale reading between 100 psi and 580 psi, for Size 1, and from +10 psi to -10 psi at any scale reading between 20 psi and 140 psi for Size 2, while connected to a pneumatic pressure source The adjustment shall contain a lock and seal to resist tampering.

3.8.1.3 Housing The gage element housing shall be of machined or extruded aluminum alloy with outside dimensions not exceeding 1-1/8 by 1-1/8 by 7-1/8 inches. A 3/4 inch diameter opening shall be provided in the housing to allow reading of the indicator. A wire marker or pointer shall be inserted at the opening midpoint to serve as a reference point to facilitate accurate reading. The opening shall be equipped with a non-breakable magnifying lens. The lens retainer or bezel shall not protrude more than 1/8 inch above the housing surface. The outer surface of the lens shall be recessed at least 1/16 inch below the surrounding surface. The housing for Size 1 gauge elements shall be anodized yellow (color 33481 of FED-STD-595) and Size 2 gage elements, blue (color 35177 of FED-STD-595) in accordance with MIL-A-8625

3.8.1.3.1 Quick-disconnect coupling The gage element housing shall be equipped with the male portion of a straight thru quick-disconnect coupling conforming to the applicable dimensions of Figure 2. The overall dimensions of the housing with the disconnect fitting shall not exceed 8-3/4 inches

3.8.1.3.2 Lens The reading port magnifying lens shall be of good commercial quality, non-breakable, and shall be free from distortion The lens shall be retained in the housing in such a manner as to enable removal for cleaning without the use of special tools. The relationship of the lens, reading marker, and indicator shall be such as to enable accurate pressure readings. The magnification of indicator markings shall be at least 2 to 1.

3.8.2 Remote-controller The remote controller shall conform to MIL-I-85352/2. The body of the remote-controller shall be of machined or extruded aluminum alloy constructed to withstand a maximum working pressure of 1,500 psi and a minimum bursting pressure of 3,000 psi. The body shall be anodized to resist corrosion The body shall be a maximum of 3-1/4 inches long by 3-1/4 inches high by 1-1/4 inches thick. The inlet port of the remote controller body shall be fitted with an AN912-2J bushing reducer. The body shall contain a lever operated inflator valve assembly and a bleed-off petcock. The attachment point

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:50Z
Check the source to verify that this is the current version before use.

for the gage element shall be equipped with the female portion of a straight thru quick-disconnect coupling to receive the gage element. This shall provide an airtight joint and be readily assembled or disassembled without the use of tools. The words "hose", "gage" and "relief valve" shall be impression stamped adjacent to the applicable port on the remote controller body.

3.8.2.1 Inflator valve assembly. The valve cartridge external envelope shall be designed and constructed in accordance with Figure 3 and shall be removable and replaceable as a unit without the use of special tools. The level or trigger mechanism shall be designed and constructed to provide a minimum leverage ration of 4.5 to 1 and shall be a depress to open design. The depressed position of the valve cartridge internal mechanism shall permit delivery of source pressure directly to the receiver. Release of the lever will position the valve cartridge mechanism so as to shut-off the pneumatic source and simultaneously permit receiver pressure to be read on the gage element. The body and valve mechanism shall be designed and constructed to prevent source pressure from being introduced into the gaging element when the lever or trigger mechanism is in any position.

3.8.2.2 Bleed-off petcock. The bleed-off petcock shall be attached to the body as indicated in Figure 1, in such manner as to provide a safe method of bleeding off excess pneumatic pressure (over-inflation) from the receiver.

3.8.2.3 Warning plate. A warning plate, manufactured from a suitable metal, shall be mechanically attached to the remote-controller body with screws or rivets. The plate shall be not less than 2-1/2 by 1-1/2 inches and shall be clearly lettered by a metal-photo or equivalent method as follows:

WARNING TO OPERATE, STAND BACK FULL LENGTH OF HOSE.
INLET PRESSURE IS TO BE SET AT 150% OF DESIRED
TIRE PRESSURE OR 600 PSI, WHICHEVER IS THE LESSER
DO NOT EXCEED 600 PSI

3.8.3 Hose/tire valve adapter assembly. The assembly shall consist of a service hose, tire valve adapter and a straight thru quick-disconnect coupling (see Figure 1). The tire valve adapter shall be firmly secured to the coupling with a 9-3/8 ± 1/2 inch length of brass chain to prevent loss. Each end of the chain shall terminate with a slip or "S" hook to allow full 360° swivel action. The end of the hose that connects to the remote controller body shall include a swivel type fitting.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

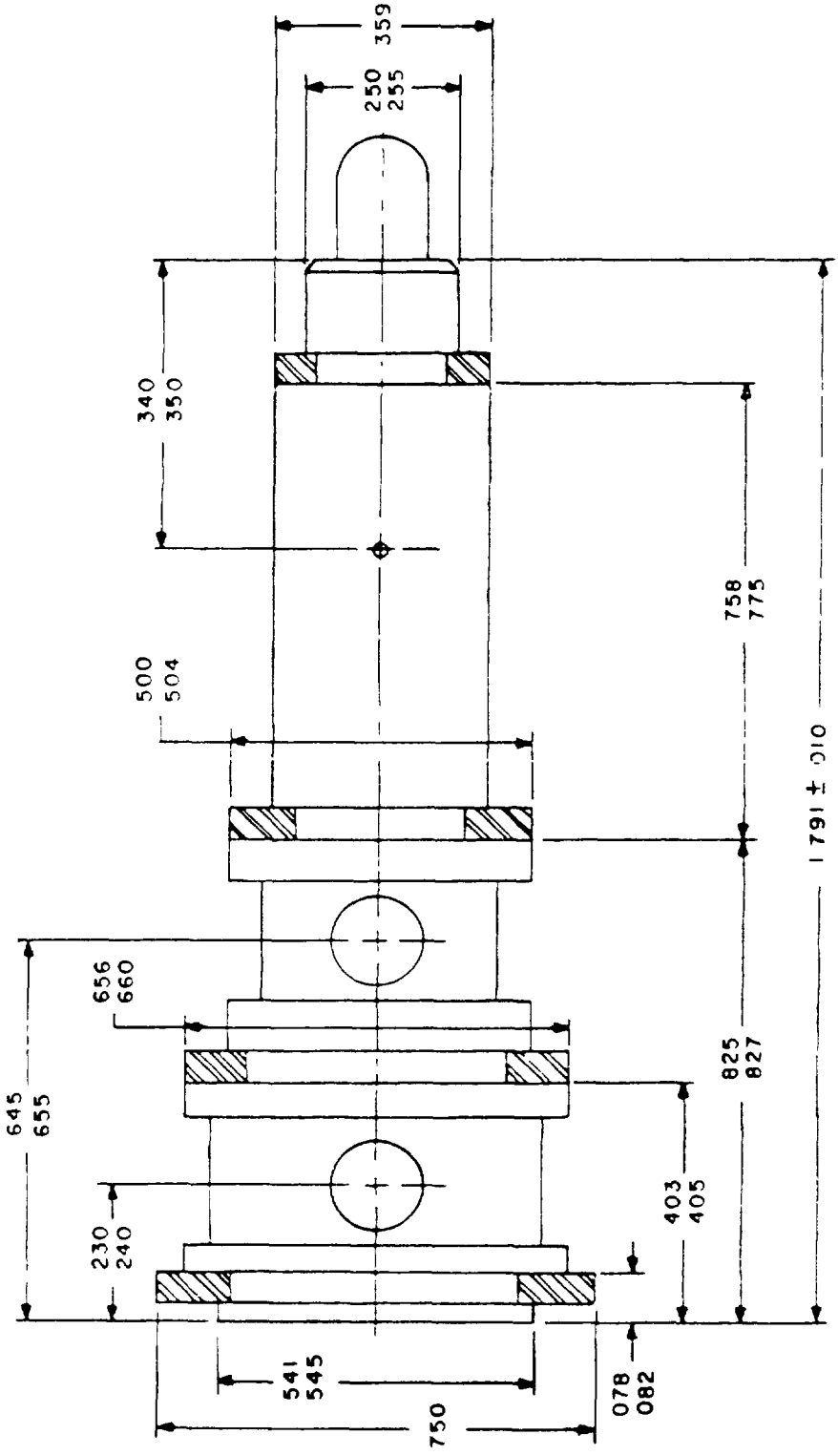


Figure 3 Valve cartridge

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

3.8.3.1 Tire valve adapter The tire valve adapter shall be fabricated of stainless steel in accordance with Figure 2. One end shall be threaded to fit a tire valve stem and contain a compression resistant gasket to create an air tight seal. The other end shall be fitted with a replaceable tire valve adapter extension. No special tools shall be required for attaching the adapter to the tire valve stem.

3.8.3.1.1 Tire valve adapter extension The tire valve adapter extension shall be so designed as to depress, without distortion or damage, the tire valve core when threaded onto the tire valve stem and connected to the quick-disconnect coupling (Figure 4). This replaceable extension is to be a non-sealing device and shall be removed and replaced in the tire valve adapter without special tools

3.8.3.2 Service hose assembly. The service hose shall conform to Type I, Class A of MIL-H-13531 The outer hose covering shall be perforated. The hose shall be 1/4 inch inside diameter and not less than 10 feet in length The end of the hose shall be fitted with a straight thru quick-disconnect coupling designed to depress the extension in the tire valve adapter (see Figure 4) The actuator pin (spud) and spring shall be designed in such a manner that the spud cannot be removed from the spring thereby eliminating the possibility of inverting the spud on the spring

3.8.4 Relief valve An adjustable relief valve shall be attached to the outlet side of the inflator body by a threaded connection as indicated in Figure 1. The relief valve shall be installed such that the vented gases are directed down and/or away from the operator The valve is to be a self reseating type and shall contain an adjustment mechanism capable of setting the relief pressure throughout a range of 100 to 650 psi The adjustment mechanism shall be graduated in psi readings so that any change in setting shall be readily detected. Once set, the valve shall be capable of repeating any relief setting throughout its range within ±5% The valve shall be capable of re-seating itself with zero leakage at not less than 85% of its relief setting The adjustment mechanism shall contain a lock and seal to resist tampering The relief valve shall be a maximum of 3-3/4 inches in length The initial setting of the relief valve shall be 100 psi.

3.8.5 Carrying case A metal case shall be provided for storing and transporting the inflator assembly The case shall incorporate two spring type latches to securely hold the lid in a closed position. The latches shall be constructed in such a manner that opening and closing of the case may be easily accomplished The case shall be provided with a handle which shall provide sufficient clearance for an easy and complete grip by a person wearing heavy gloves. The case shall

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

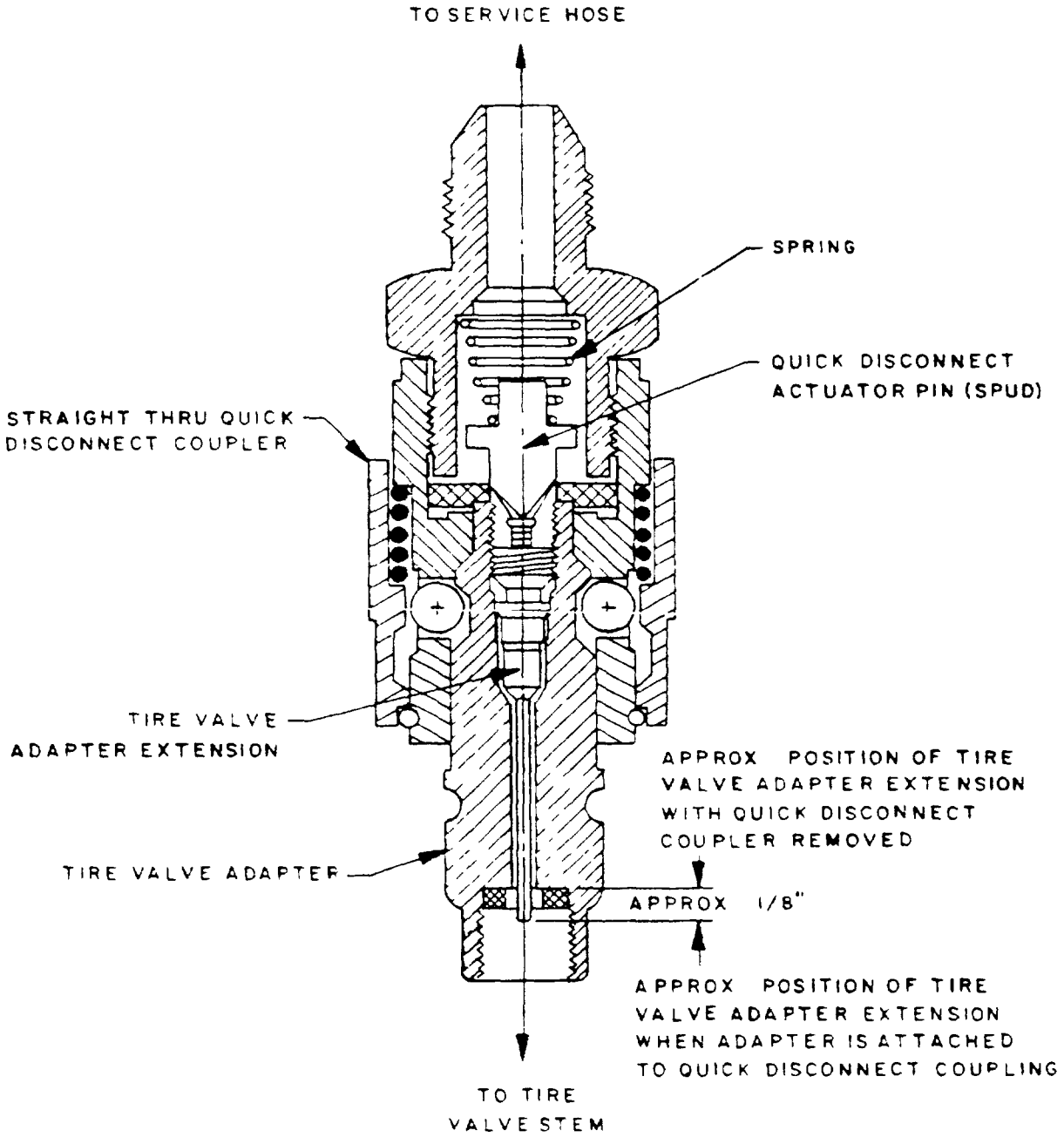


Figure 4. Tire valve adapter coupling assembly

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

be fitted with internal cushioning, utilizing rubberized hair conforming to type II of PPP-C-1120, to prevent the component parts of the inflator assembly from shifting position in the case and protects same from damage during storage and transporting. The case shall incorporate an instruction sheet in accordance with 3.7.5.1. The outside dimensions of the closed case shall not exceed 18 inches long by 12 inches wide by 4 inches high (see Figure 5). The carrying case shall be finished in accordance with MIL-STD-808, FIO3 Type I, Color DG.

3.8.5.1 Operating instructions. An instruction sheet covering the care and safe operation of the inflator shall be permanently attached to the inside of the cover (see Figure 5). The instruction shall be enclosed in a transparent plastic envelope, permanently sealed on all edges, or printed on a decalcomania. The envelope or decalcomania shall be permanently fastened to the cover in such manner that the instructions can be read from the front of the open case.

3.8.6 Inlet fitting. An inlet fitting shall be fabricated in accordance with Figure 6. It shall be installed in the remote-controller as shown in Figure 1.

3.9 Dual chuck stem gage kit. The dual chuck stem gage kit shall conform to MIL-I-85352/4 and shall consist of a dual-chuck foot, chuck foot extension, a gage element coupled together with a quick-disconnect coupling and carrying case.

3.9.1 Dual-chuck foot. The chucks shall be offset approximately 30 degrees as shown on Figure 7. The chucks shall be designed and constructed to fit a valve stem conforming to MS28889. The dual chucks shall be so designed that engagement of one of the chucks with the valve stem will render the deflator stem of the other chuck inoperative. Deflator pins shall be fabricated from hardened steel. The pins shall be designed with flats so that distortion of sealing washers will not cause any restriction of the air flow through the chuck to the gage after continued usage.

3.9.2 Chuck foot extension. The chuck foot shall have an extension approximately 6 inches long. The chuck foot shall either be fabricated integrally with the extension or shall be rigidly attached thereto in such manner that it will not become loose or detached under normal usage. The free end shall be equipped with the female portion of a quick disconnect coupling which will mate with the male portion of a quick-disconnect coupling conforming to the applicable dimensions of Figure 2.

3.9.3 Gage element. The gage element shall conform to MIL-I-85352/3 and meet all the requirements specified in 3.7.1, 3.7.1.1, 3.7.1.2, 3.7.1.3, 3.7.1.3.1 and 3.7.1.3.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

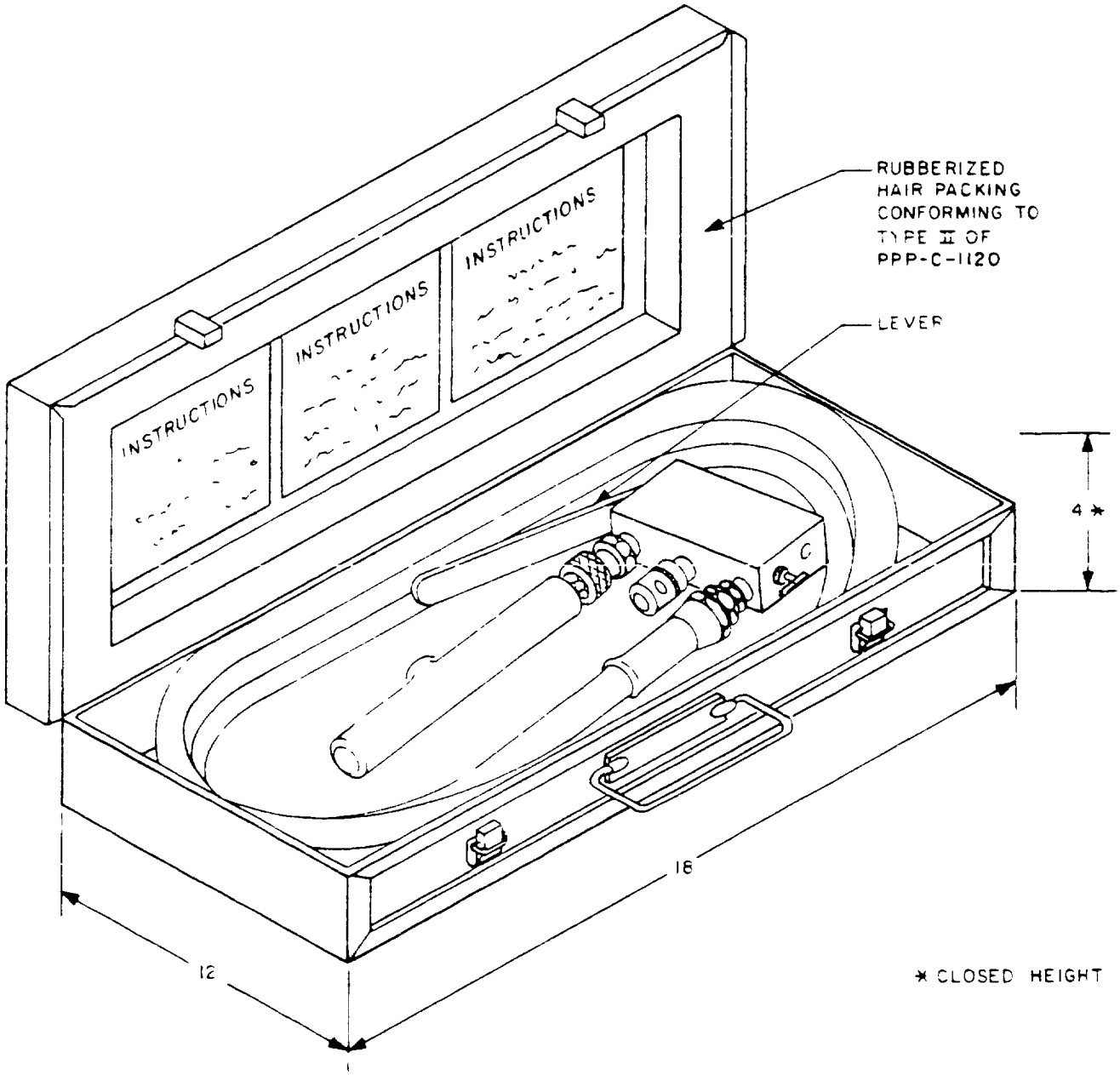


Figure 5 Inflator assembly carrying case

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

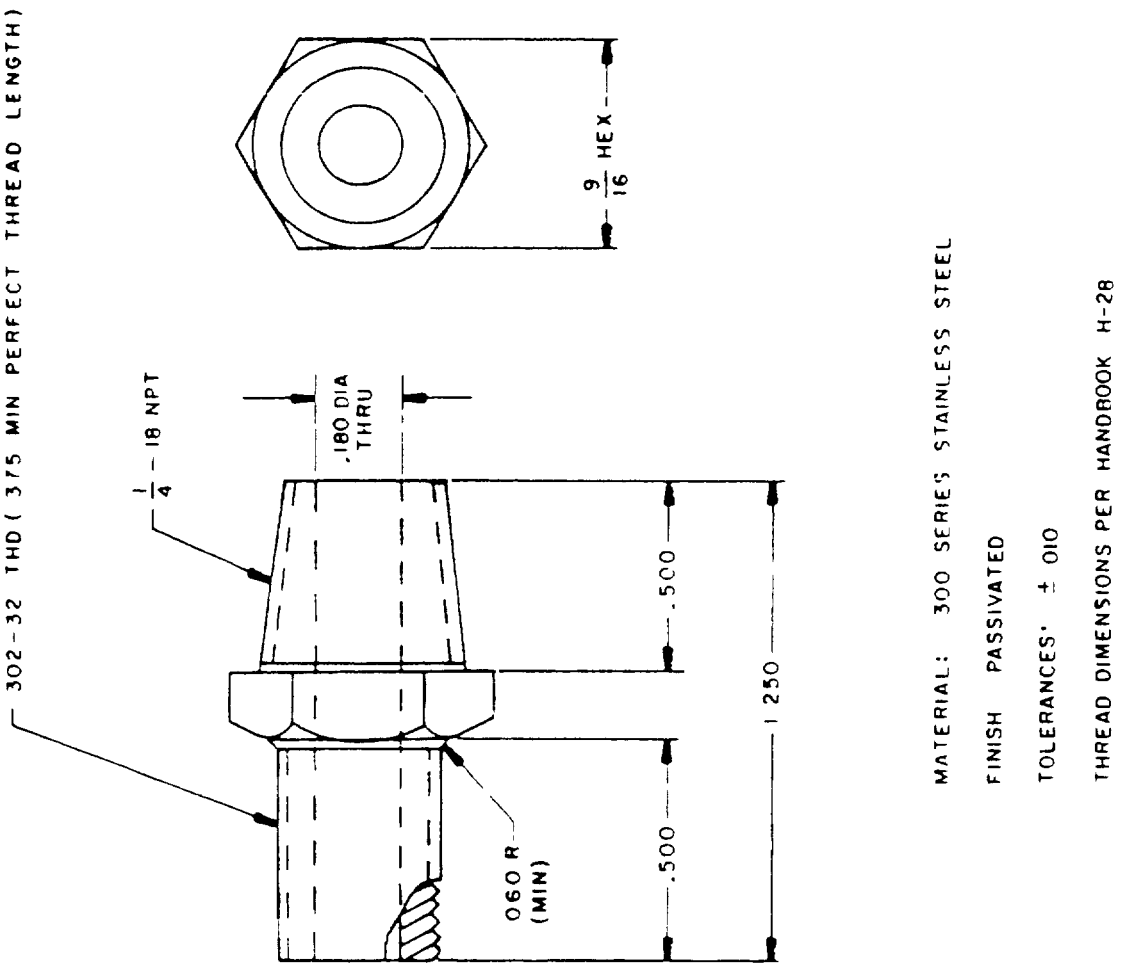


Figure 6 Inlet fitting

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:50Z
Check the source to verify that this is the current version before use.

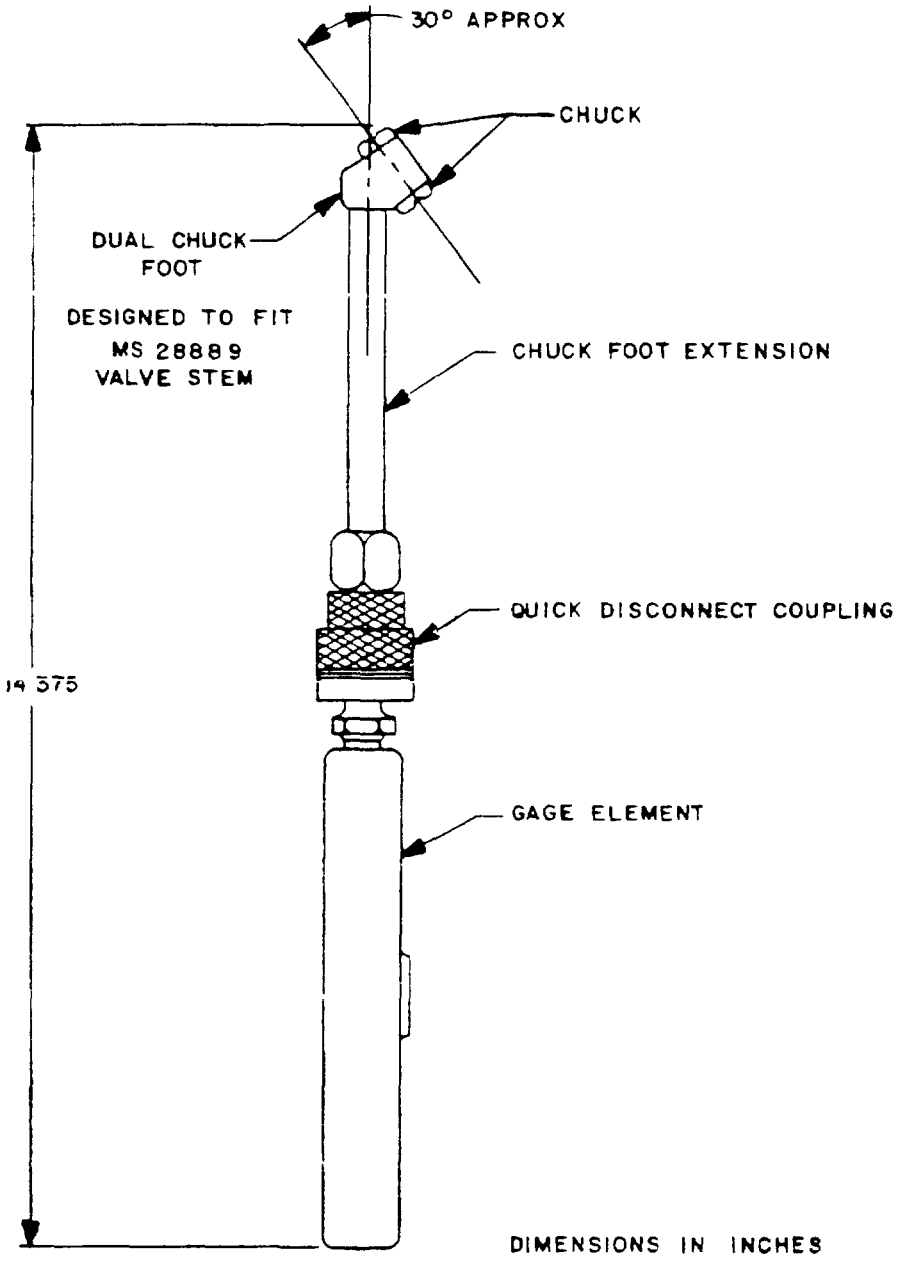


Fig re 7 Dual chuck stem gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

3.9.4 Dual chuck gage carrying case A metal case shall be provided for storing and transporting the dual chuck stem gage. The case shall incorporate two spring type latches to securely hold the lid in a closed position. The latches shall be constructed in such a manner that opening and closing of the case may be easily accomplished The case shall be provided with a handle which shall provide sufficient clearance for an easy and complete grip by a person wearing heavy gloves The case shall be fitted with internal cushioning, utilizing rubberized hair conforming to type II of PPP-C-1120, to prevent the dual chuck stem gage from shifting position in the case and protects same from damage during storage and transporting. The outside dimensions of the closed case shall not exceed 10-5/8 inches long by 5 inches wide. The carrying case shall be finished in accordance with MIL-STD-808, F103 Type I, Color DG

3.10 Part numbering of interchangeable parts All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable The item identification and part number requirements of MIL-D-1000/1 shall govern the manufacturer's part numbers and changes thereto

3.11 Protective finish Materials, used in the construction of the inflator assembly, that are subjected to deterioration when exposed to environmental conditions likely to occur during service usage shall have a protective finish to resist corrosion The use of any protective coating that will crack chip, or scale with age or extremes of atmospheric conditions shall be avoided. Protective finishes shall be in accordance with QQ-N-290, QQ-P-416, QQ-Z-325, MIL-A-8625 or QQ-C-320. When used, cadmium plating shall be type II or III in accordance with QQ-P-416, and of a class that is adequate to achieve the degree of protection required

3.12 Identification marking Equipment assemblies, and parts shall be marked for identification in accordance with MIL-STD-130 The nomenclature and scale range of both gage elements shall be marked on the outside of the carrying case in black letters 3/8 inch high

3.12.1 Each gage element is to have separate identification. The gage element housing shall be marked on the indicator side of the housing with the proper pressure range, 50 to 600 psi for Size 1, and 10 to 150 psi for Size 2, as follows

GAGE PRESSURE
XX TO XXX PSI

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

On the side of the housing adjacent to the above, the following shall be marked:

U S MILITARY PROPERTY
N.S.N
MFR
CONT NO
P/N M85352/3-X (X-insert correct dash number)

These markings shall be letters 1/8 inch high, stamped or etched

3.12.1.1 Calibration tag. A pressure sensitive metallic foil tag approximately 3/4 inch square shall be firmly attached to the indicator side of the gage element as indicated in Figure 1. The tag shall indicate that the gage element has been tested and is in compliance with the accuracy requirements of MIL-I-85352. The date of the test shall be included.

3.12.2 Remote controller assembly. On the side of the remote controller housing, opposite the warning marking, the following shall be marked:

U S MILITARY PROPERTY
N.S.N
MFR
CONT NO
P/N M85352/2

3.13 Cleaning. The inflator assembly and dual chuck stem gage shall be thoroughly clean of loose, spattered or excess solder, metal chips, or other foreign material. There shall be no burrs and sharp edges, as well as resin flash which may crumble.

3.14 Workmanship. Workmanship shall be of the quality necessary to produce complete units of good appearance, with safe and efficient operating characteristics as is customarily found in high grade commercial quality items of a similar nature.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

4.2 Classification of inspection The inspection of the inflator assembly shall be classified as follows.

a First article inspection (4.3)

b. Quality conformance inspection (4.4)

4.3 First article inspection. First article inspection shall consist of the inspections of this specification.

4.3.1 Sampling for first article inspection Unless otherwise specified, as soon as practicable after award of the contract or order, the contractor shall furnish 2 inflator assembly kits or 2 remote controller assemblies or 2 gage elements, or 2 dual chuck stem gages as applicable, to determine conformance with this specification The samples shall be identified in accordance with 3.10 and contract number and shall be inspected as specified herein and in accordance with the contract or order (see 6.2)

4.3.2 First article test report The contractor shall prepare a report of the first article tests in accordance with MIL-STD-831 showing that the inflator assemblies for each size gage element conform to this specification. Two copies shall be furnished to the procuring activity Two additional copies shall be forwarded with the verification test samples.

4.3.3 Verification test samples for the procuring activity The contractor shall submit to the procuring activity, together with the first article test report, 2 sample inflator assembly kits or remote controller assemblies or gage elements or dual chuck stem gages as applicable, as specified in 4.3.1 to be used as follows:

a For a review of the mechanical construction of the product

b. To perform any tests included in the specification, after reviewing the contractor's test report.

4.4 Quality conformance inspection Quality conformance inspection shall consist of the individual inspection and sampling tests

4.4.1 Individual inspection Each test sample shall be subjected to the following inspection

a Examination (4.6.1)

b. Scale error at room temperature (4.6.2)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

4.4.2 Sampling tests. A random sample shall be selected from each inspection lot in accordance with MIL-STD-105, Inspection Level S-2, Acceptable Quality Level (AQL) of 0.65 percent major defects and an AQL of 10.0 percent for minor defects. Samples shall be subjected to the tests specified in 4.6.1, 4.6.2, 4.6.3 and 4.7.

4.4.2.1 Inspection lot. A lot shall consist of all inflator assemblies and carrying cases or remote controllers or gage elements or dual chuck stem gages manufactured and assembled under essentially the same conditions and submitted for inspection at one time.

4.4.2.2 Resubmitted inspection lots. The paragraph entitled "Resubmitted lots or batches" of MIL-STD-105 shall apply except that a resubmitted inspection lot shall be inspected by the contractor under supervision of the Government inspector, using tightened inspection. If the original acceptance number was zero, a sample size represented by the next higher sample size code letter shall be chosen with the acceptance number remaining zero. Before an inspection lot is resubmitted, full particulars concerning the cause of previous rejection and the action taken to correct the defects in the inspection lot shall be furnished by the contractor to the Government inspector.

4.5 Test conditions.

4.5.1 Test equipment. Tests shall be conducted using equipment of laboratory accuracy and acceptable to the procuring activity. Tests shall be performed using air or nitrogen gas pressure to simulate actual use. Tests with hydraulic pressures are not acceptable.

4.5.2 Atmospheric conditions. Unless otherwise specified, all tests required by this specification shall be made at an atmospheric pressure of approximately 30 inches of mercury and at room temeprature of $70^{\circ} \pm 5^{\circ}$F. When tests are made with atmospheric pressure and temperature substantially different from the above values, proper allowance shall be made for the difference from specified conditions.

4.6 Inspection methods.

4.6.1 Examination. The inflator assembly and dual chuck stem gage including carrying case, shall be thoroughly examined for design, construction, materials, and workmanship to determine compliance with the requirements of this specification. Any deviation from these requirements shall be cause for rejection.

4.6.2 Scale error at room temperature. The inflator assembly shall be tested for scale errors at 1/4, 1/2, 3/4, and full-range pressure minus 20 psig of the gage element. The test shall be made by subjecting the inflator assemblies to the pressures specified

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:50Z
Check the source to verify that this is the current version before use.

herein to produce these readings, first with the pressure increasing, then with the pressure decreasing. With the pressures increasing, the pressure shall be brought to, but shall not exceed, the pressure specified to give the desired reading, and with the pressure decreasing, the pressure shall be brought down to, but shall not fall below, the pressure specified to give the desired reading. Accuracy of Size 1 gages at each reading shall be within ±6 psig. Accuracy of Size 2 gages at each reading shall be within ±2 psig from 10 to 80 psig and ±4 psi from 81 to 150 psig. The inflator assembly test shall be conducted on a test set-up as shown on Figure 8. The receiver test gage, shown in this test set-up, shall conform to the specifications of ANSI B40.1-1974 for test gages It shall have a pressure range of not more than 750 psi, and shall be graduated in increments not larger than 2 psig.

4.6.3 Overpressure With the relief valve removed and its attachment point plugged to prevent leakage, the inflator assembly with lever depressed shall be subjected to a pressure of 3000 psig. The lever shall be released and the inflator assembly shall be subjected to the above test pressure for a period of 10 minutes. The gage element and remote controller assembly shall show no sign of leakage or rupture. Not less than 5 minutes following the application of the specified pressure, the inflator assembly shall be subjected to the test of 4.6.2 No calibration or repair of the gaging elements shall be permitted. If the accuracy exceeds the tolerance specified, the gaging elements shall be rejected. This test shall be performed with both size gage elements.

4.6.4 Endurance The inflator assembly and dual chuck stem gage shall be subjected to 10,000 applications of pressure from 0 to 80 ± 5 percent of full-scale pressure and return to 0 at the rate of 10 ± 5 cpm. The applied pressure with respect to time shall approximate a sine wave shape Not more than 5 minutes following this test, the inflator assembly and dual chuck stem gage element shall be subjected to the test of 4.6.2 No recalibration or repair of the gaging elements shall be permitted If the accuracy exceeds the tolerance specified, the gaging elements shall be rejected.

4.6.4.1 Functional test. The inflator assembly functional test shall be conducted on the test set-up shown in Figure 8 Pneumatic pressure to the remote-controller shall be regulated to 1500 psi The inflator valve shall be slowly depressed to allow gas to flow into the receiver. There shall be absolutely no indication of pressure on the gage element as the inflator valve lever is depressed The gage element indicator must remain absolutely stationary while gas is flowing into the receiver. At the end of a 5 second interval, the inflator valve lever shall be slowly released. The valve shall instantly stop the flow of gas to the receiver. The gage element indicator shall indicate the pressure in the receiver to within ±6 psi for the Size 1 gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

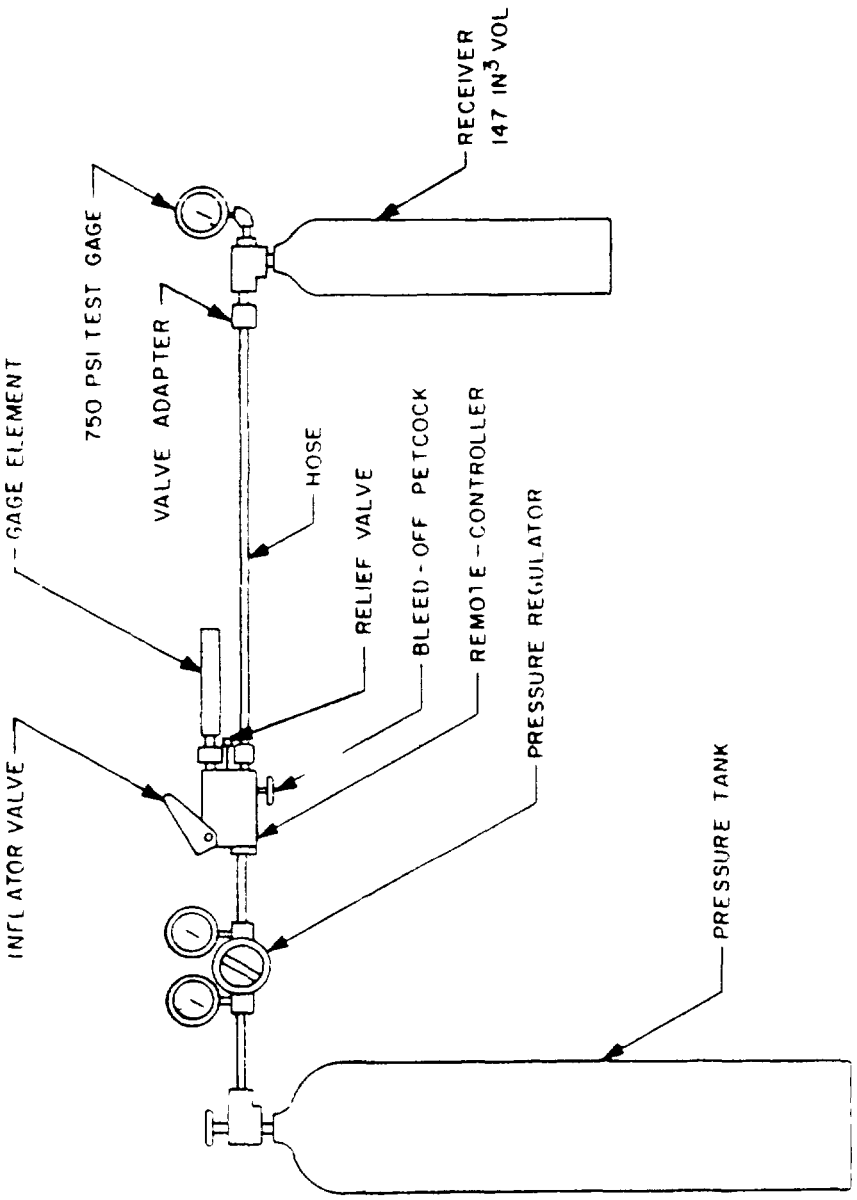


Figure 8. Inflator assembly task set-up

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

element, and ±2 psig from 10 to 80 psig and ±4 psig from 81 to 150 psig for the Size 2 gage element without overshoot. The inflator valve lever shall be slowly depressed and released a total of 4 times to bring the receiver pressure up to the gage element capacity. Any evidence of the source pressure being introduced at the gage element or delay in the valve instantly shutting off the flow of the source gas shall be cause for rejection.

4.6.4.2 Relief valve test. The relief valve test shall be conducted on the test set-up shown in Figure 8 with source pressure regulated to 1500 psi. The relief valve shall be adjusted to vent at 100 psi. The inflator lever shall be depressed to pressurize the receiver until the relief valve opens. The relief valve shall prevent over-inflation. The test gage reading shall not exceed the relief valve setting by more than 5%. The bleed-off petcock shall be opened to reduce receiver pressure to 85% of relief valve setting. The relief valve shall reseat and exhibit zero leakage. The above tests shall be repeated for a total of four times for this and for relief valve settings of 200, 350, 500 and 650 psi.

4.6.5 Drop test. The inflator assembly and dual chuck stem gage shall be dropped at least 25 times from a height of not less than 4 feet on a concrete floor. The inflator assembly and dual chuck stem gage shall strike the floor in various attitudes, i.e. the valve pressure gage, chuck etc. shall hit the floor rather than the hose. The lens of the pressure gage shall not be chipped or cracked and the gage performance shall not be impaired. After the drop test, the gage elements shall be subjected to the test specified in 4.6.2. No recalibration or repair of the gage elements shall be permitted. If the accuracy exceeds the tolerance specified, the gage element shall be rejected.

4.6.6 Cold test. Gage elements shall be subjected to a cold temperature environment of -20°F for not less than 6 hours. Not more than 3 minutes after removal from the cold temperature environment, the gage elements shall be subjected to the test specified in 4.6.2. No recalibration or repair of the gage elements shall be permitted. If the accuracy exceeds the tolerance specified, the gage elements shall be rejected.

4.6.7 Heat test. Gage elements shall be subjected to a heat temperature environment of 140°F for not less than 6 hours. Not more than 3 minutes after removal from the hot temperature environment the gage elements shall be subjected to the test specified in 4.6.2. No recalibration or repair of the gaging elements shall be permitted. If the accuracy exceeds the tolerance specified, the gage elements shall be rejected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

4.6.8 Friction test. Gage elements shall be tested for inherent friction on the test set-up as shown in Figure 8. The gage shall be tested at no less than four points over the scale range. For Size 1, the pressure in the receiver shall be brought up to approximately 110 psi by depressing the inflator valve on the remote-controller. The lever shall be released and the bleed-off petcock shall be opened to allow pressure in the receiver to bleed down at not less than 10 nor more than 20 psi per minute. The bleed-off petcock shall be shut off as soon as the movement of the indicator in the gage element is noted. The difference in the pressures indicated by the gage element indicator and the test gage shall not exceed 30 psi. The test shall be repeated at approximately 300 psi, 450 psi and 600 psi. For Size 2, the pressure in the receiver shall be brought up to approximately 28 psi by depressing the inflator valve on the remote-controller. The lever shall be released and the bleed-off petcock shall be opened to allow pressure in the receiver to bleed down at not less than 10 or more than 20 psi per minute. The bleed-off petcock shall be shut off as soon as the movement of the indicator in the gage element is noted. The difference in the pressures indicated by the gage element indicator and the test gage shall not exceed 10 psi. The test shall be repeated at approximately 75 psi, 112 psi, and 150 psi. A total of 10 tests shall be conducted, each test consisting of an examination at four points. Any evidence of failure to indicate pressure with the accuracy requirements stated above shall be cause for rejection.

4.6.9 Calibration test. The calibration test shall be conducted on the test setup shown on Figure 8. The pressure in the receiver shall be brought up to half the gage element range. For Size 1, the adjustment mechanism of the gage element shall be adjusted to indicate 20 psi above the reading on the receiver gage, and then readjusted to indicate 50 psi below the reading on the receiver gage. For Size 2, the adjustment mechanism of the gage element shall be adjusted to indicate 10 psi above the reading on the receiver gage, and then readjusted to indicate 10 psi below the reading on the receiver gage. The adjustment mechanism for Size 1 and Size 2 gages shall then be adjusted until the indicator matches the reading on the receiver gage. The adjustment mechanism shall be locked and the gage subjected to the accuracy test of 4.6.2. Three additional adjustments are permitted to calibrate the gage to meet the accuracy requirements of 4.6.2.

4.6.10 Valve adapter test. It shall be demonstrated that the tire valve adapter can be applied to a standard tire valve stem and valve stem extension without use of special tools. After installation, the quick-disconnect coupling shall be engaged and disengaged for 1000 engagements. During this test, after each 100 engagements, the tire valve core and the tire valve adapter extension shall be removed.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:50Z
Check the source to verify that this is the current version before use.

and examined. There shall be no deflection or evidence of failure in either component. The tire valve adapter shall be installed a total of 100 times on a standard tire valve stem. The valve adapter shall be disassembled and examined for wear or compression of the gasket. After completion of 1000 engagements and 100 installations the assembly shall be tested to withstand an integral pressure of 750 psi for 10 minutes During this time there shall be no indication of leakage.

4.7 Packaging, packing, and marking Preparation for delivery shall be examined for conformance to section 5.

5. PACKAGING

5 1 Preservation and packaging Packaging shall be level A or C in accordance with MIL-STD-794. Preservation shall be Method II in accordance with MIL-P-116 (see 6 2)

5.1 1 Unit packaging The unit packaging shall be considered as the carrying case complete with the inflator assembly therein. The metal carrying case shall be utilized as the interior container.

5 2 Packing Packing shall be level A, B, or C in accordance with MIL-STD-794 (see 6 2)

5 3 Marking of shipments. In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with MIL-STD-129 (see 6.2).

6. NOTES

6 1 Intended use The inflator assembly covered by this specification is intended for the measurement of pneumatic pressure when inflating aircraft tires or any pneumatic receiver having internal pressure from 10 to 600 psi The dual chuck stem gage is intended for measuring pneumatic pressure in aircraft tires or pneumatic receivers from 10 to 600 psi The inflator assembly and dual chuck stem gage is intended for use at both land base and ship base locations.

6 2 Information for contracting officer. Contracts or orders should specify the following

a. Title, number and date of this specification

b Selection of applicable levels of preservation packaging, and packing required (see 5.1 and 5 2)

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:50Z
Check the source to verify that this is the currentversion before use.

c. Special marking, if required (see 5.3).

d. The inflator assemblies, gage elements, dual chuck stem gages or combinations of these should be ordered by part number as specified, see specification sheets 1, 2, 3 and 4

e. Where the first article test samples should be sent and instructions concerning the submittal of the test reports (see 4.3 3).

f. For Navy procurement

(1) Maintenance instructions and parts list required by para 6 3 shall be submitted to the Commanding Officer, Naval Air Engineering Center (Attn Code 92713) Lakehurst, NJ 08733 for approval.

(2) Engineering change procedure in accordance with NAVAIR INST 4275.3A The Naval Air Systems Command Department of the Navy, Washington, DC 20360. (Attn AIR-534) will be responsible for review of Class I changes under the contract. The Commanding Officer, Naval Air Engineering Center, Lakehurst, NJ 08733 (Attn: Code 92713) will be responsible for approval of Class II changes

6 2.1 Contracts or orders should specify the following provisions for first article inspection.

6.2.1.1 Whether first article inspection is required When a contractor is in continuous production of the inflator assembly or dual chuck stem gages from contract to contract, consideration should be given to waive the first article inspections. If first article inspection is required, indicate

a. Where the first article inspection is to be conducted (at the contractor's plant or Government or commercial laboratory)

b. That the approval of first article samples or the waiving of the first article inspection shall not relieve the contractor of his obligation to fulfill all other requirements of the specification and contract.

6 3 Maintenance instructions and parts list. A loose leaf sheet or sheets shall be provided with each inflator assembly delineating instructions for the calibration of gage elements, adjustment of the pressure relief valve and any other maintenance functions In addition, the sheet/sheets shall contain a complete illustrated parts breakdown for the entire inflator assembly including the vendor/subvendor's identification and the vendor's/subvendor's part numbers for all components.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

The sheets shall be packed inside the carrying case and shall be protected with a plastic envelope to prevent tearing or similar damage during shipment or storage. The envelope shall be permanently affixed to the inside bottom of the case. For Navy procurement the instruction and parts list shall be subject to the approval of the Commanding Officer, Naval Air Engineering Center (Attn: 92713).

Custodian
Air Force - 99
Navy - AS
Army - AV

Review activity:
DLA - GS

Preparing Activity
Navy - AS
(DoD Project No. 6685-0634)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
Naval Air Engineering Center
Lakehurst, NJ 08733

POSTAGE AND FEES PAID
DEPARTMENT OF THE NAVY
DoD 316



OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Engineering Specifications and Standards Department (Code 93)
Naval Air Engineering Center
Lakehurst, NJ 08733

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS This form is provided to solicit beneficial comments which may improve this document and enhance its use DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government Fold on lines on reverse side, staple in corner, and send to preparing activity Attach any pertinent data which may be of use in improving this document If there are additional papers, attach to form and place both in an envelope addressed to preparing activity A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed

NOTE This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements

DOCUMENT IDENTIFIER (Number) AND TITLE MIL-I-85352 INFLATOR ASSEMBLY AND GAGE ELEMENTS, PNEUMATIC PRESSURE, REMOTE CONTROL, DIRECT READING

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1 ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID RESTRICTIVE LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDED WORDING CHANGE

C REASON FOR RECOMMENDED CHANGE(S)

2 REMARKS

SUBMITTED BY (Printed or typed name and address — Optional)

TELEPHONE NO

DATE

DD FORM 1 OCT 76 1426 Replaces edition of 1 Jan 72 which may be used S/N 0102-LF-001-4260

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:50Z
Check the source to verify that this is the current version before use.